Lecture 12: Regression Inference

From Fitting Lines to Testing Coefficients

Previously, on Lectures 10 & 11...

t-tests: Compare one or two means. Welch's for unequal variances.

χ^2: Goodness-of-fit and independence for counts/categories.

ANOVA: Compare k group means via $F = \mathrm{MS_B}/\mathrm{MS_W}$.

A/B testing: Same hypothesis testing, different engineering (peeking, MDE).

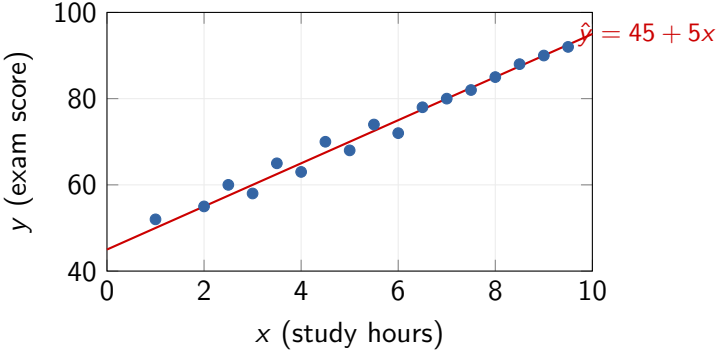
All tests: Assume some model, measure evidence against H_0.

Today: The most widely used model in statistics: **linear regression**.
Not just fitting a line — testing *which predictors matter* and *how much*.

Part I: The Linear Model

From the math module to the statistical framework

Quick Recall: OLS Regression



Math module: Find $\hat{\beta}_0, \hat{\beta}_1$ that minimize $\sum(y_i - \hat{y}_i)^2$.

Statistics question: Is β_1 *really* nonzero, or could the slope be noise?

The Statistical Model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i$$

$$\text{where } \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

$\beta_0, \beta_1, \ldots, \beta_p$
True (unknown) population parameters. *Fixed* numbers.

ε_i
Random error. Everything the model doesn't capture.

$\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$
OLS estimates. *Random* — they vary from sample to sample.

Key shift: $\hat{\beta}$ has a *sampling distribution*, just like $\bar{X}$ from L7.

Assumptions of the Linear Model

1. **Linearity:** $E[Y \mid X] = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$.
   Check: residuals vs fitted plot should show no pattern.

2. **Independence:** Errors ε_i are independent of each other.
   Violated by: time series, clustered data, spatial data.

3. **Homoscedasticity:** $\mathrm{Var}(\varepsilon_i) = \sigma^2$ (constant variance).
   Check: residuals vs fitted — fan shape = heteroscedasticity.

4. **Normality:** $\varepsilon_i \sim N(0, \sigma^2)$.
   Check: QQ plot of residuals. Least important for large n (CLT helps).

Mnemonic: **LINE** — Linearity, Independence, Normality, Equal variance.

Part II: Inference for Coefficients

SE, t-tests, p-values, and CIs for each β_j

The Sampling Distribution of $\hat{\beta}$

OLS estimates are random variables with known distributions

Under the linear model assumptions:

$$\hat{\beta}_j \quad \sim \quad N\left(\beta_j,\ \sigma^2 (X^\top X)^{-1}_{jj}\right)$$

$\hat{\beta}_j$ is **unbiased**: $E[\hat{\beta}_j] \quad = \quad \beta_j$. Variance depends on σ^2 and the design matrix X.

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Standard error of $\hat{\beta}_j$: $\mathrm{SE}(\hat{\beta}_j) \;=\; \hat{\sigma}\sqrt{(X^\top X)^{-1}_{jj}}$

where $\hat{\sigma}^2 \;=\; \dfrac{\sum e_i^2}{n-p-1} \;=\; \dfrac{\mathrm{RSS}}{n-p-1}$ (residual standard error)

This is just SE = (estimated SD) / (effective $\sqrt{n}$), same idea as L7.

Simple Linear Regression: SE Formula

For $Y = \beta_0 + \beta_1 X + \varepsilon$, the formulas simplify

$$\begin{aligned} \mathrm{SE}(\hat{\beta}_1) &= \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}} \\ \hat{\sigma} &= \sqrt{\mathrm{RSS}/(n-2)} \end{aligned}$$

SE is small when:

- Residuals are small ($\hat{\sigma}$ small)
- X's are spread out
- Sample size n is large

SE is large when:

- Lots of noise ($\hat{\sigma}$ large)
- X's are clustered together
- Sample size n is small

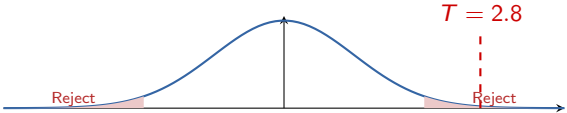
Intuition: a slope is hard to estimate when X doesn't vary much (imagine all points at $X = 5$).

t-Test for a Single Coefficient

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0$$

$$T = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1}$$

under H_0



Interpretation: "Is this predictor useful, *controlling for* all others in the model?"
$H_0 : \beta_j = 0$ means X_j has no linear effect on Y after accounting for other predictors.

Example: Predicting House Prices

$n = 100$ houses. Model: $\text{Price} = \beta_0 + \beta_1 \cdot \text{Size} + \beta_2 \cdot \text{Bedrooms} + \beta_3 \cdot \text{Age} + \varepsilon$.

	Estimate	**SE**	t	p**-value**
(Intercept)	50.2	15.3	3.28	0.001
Size (sq ft)	0.12	0.015	8.00	< 0.001
Bedrooms	5.1	4.8	1.06	0.291
Age (years)	-0.85	0.22	-3.86	< 0.001

Size: $t = 8.0$, $p < 0.001$
Strong evidence. Each sq ft adds \$120 (significant).

Bedrooms: $t = 1.06$, $p = 0.29$
Not significant. After controlling for size, bedrooms don't add much info.

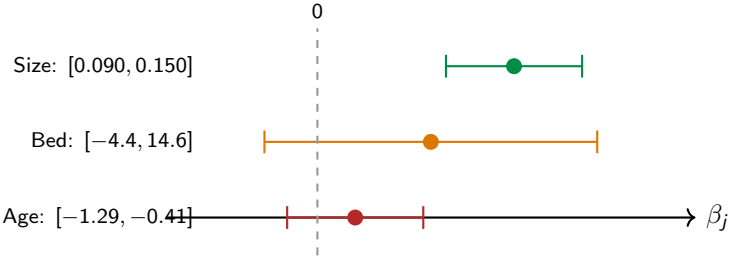
Age: $t = -3.86$, $p < 0.001$
Significant. Each year reduces price by \$850.

Confidence Intervals for Coefficients

$$\hat{\beta}_j \quad \pm \quad t_{n-p-1,\,\alpha/2} \cdot \mathrm{SE}(\hat{\beta}_j)$$

Same Wald CI formula from L8, applied to regression coefficients.

House price example (95% CI, $t_{96,0.025} \approx 1.985$):



Bedrooms CI contains 0 ⇒ not significant. Size and Age CIs exclude 0 ⇒ significant.
CI and p-value always agree: CI contains 0 ⇔ $p > \alpha$.

Part III: Overall Model Fit

Is the model as a whole useful? How much does it explain?

The Overall F-Test

Tests whether *any* predictor is useful

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{(no predictor matters)}$$

$$F = \frac{(\mathrm{SS_T} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} = \frac{\mathrm{MS_{Reg}}}{\mathrm{MS_{Res}}} \sim F_{p,\, n-p-1}$$

This is exactly ANOVA applied to regression!

- $\mathrm{SS_{Total}}$ = total variation in Y
- $\mathrm{SS_{Regression}} = \mathrm{SS_T} - \mathrm{RSS}$ = variation explained by the model
- RSS = residual sum of squares = unexplained variation

Same logic: if the model explains much more than noise, F is large $\Rightarrow$ reject H_0.

House prices: $F = 85.3$ on 3 and 96 df, $p < 0.001$. The model is highly significant.

R^2 and Adjusted R^2

$$R^2 = 1 - \frac{\text{RSS}}{\text{SS}_\text{T}}$$

Proportion of variance in Y explained by model.

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{SS}_\text{T}/(n-1)}$$

Penalizes for adding useless predictors.

$R^2 < 0.3$	$R^2 \approx 0.5$	$R^2 \approx 0.7$	$R^2 > 0.9$
Weak	**Moderate**	**Good**	**Excellent**

Trap: R^2 *always* increases when you add predictors, even useless ones.
Use R^2_{adj} to compare models with different numbers of predictors.
"I got $R^2 = 0.95$" means nothing if you have $n = 20$ and $p = 18$.

Part IV: Why OLS?

The Gauss–Markov theorem: OLS is the best you can do (with conditions)

The Gauss–Markov Theorem

Gauss–Markov Theorem

Under the assumptions of linearity, independence, homoscedasticity, and $E[\varepsilon] = 0$ (normality **not** required!),

OLS is **BLUE**: Best Linear Unbiased Estimator.

Among all *linear, unbiased* estimators of β, OLS has the *smallest variance*.

Best smallest variance	**L**inear $\hat{\beta} = AY$ for some matrix A	**U**nbiased $E[\hat{\beta}] = \beta$	**E**stimator it's a function of the data

Connects to L3: among unbiased estimators, lowest variance = best. Same principle as Cramér–Rao.

When Gauss–Markov Doesn't Help

Heteroscedasticity: Variance of ε changes with X.
Fix: Use **robust (HC) standard errors** — same $\hat{\beta}$, different SE. `statsmodels: .get_robustcov_results('HC3')`

Non-linearity: True relationship is curved, not a line.
Fix: Transform X (log, polynomial) or use nonlinear models.

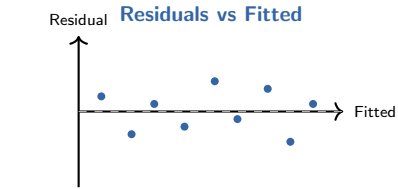
Biased estimators can win: OLS is best among *unbiased*, but ridge/lasso (from L6) trade a little bias for much less variance.
Fix: Regularization when p is large or predictors are correlated.

Gauss–Markov says OLS is the best *unbiased linear* estimator.
It does **not** say OLS is the best estimator, period. (Recall L3: bias-variance tradeoff.)

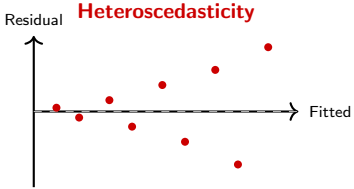
Part V: Diagnostics

How to check if your model's assumptions hold

The Four Diagnostic Plots

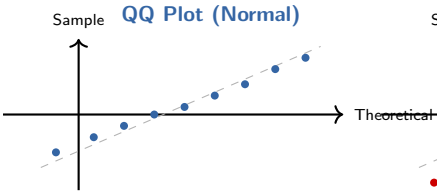


✓ Random scatter = good

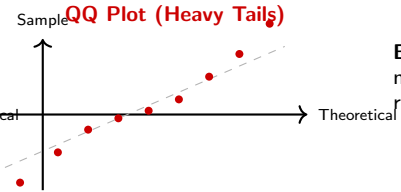


✗ Fan shape = unequal variance

Top: Check linearity & homoscedasticity



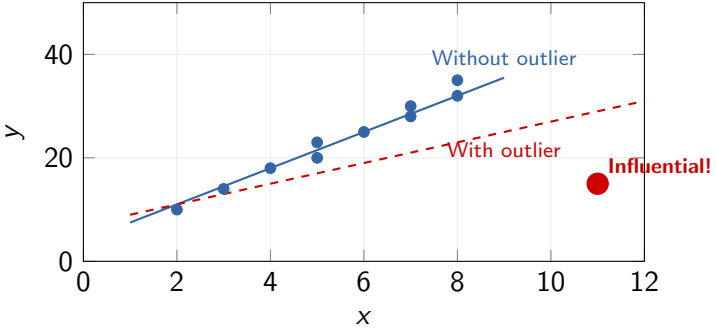
✓ Points on line = normal



✗ S-shape = non-normal tails

Bottom: Check normality of residuals

Leverage and Influential Points



Leverage (h_{ii}): How far X_i is from $\bar{X}$. High leverage = *potential* to influence the fit. Rule: $h_{ii} > 2(p+1)/n$ is high.

Cook's distance (D_i): How much the fit changes if you remove point i. Rule: $D_i > 1$ is influential.

Part VI: Logistic Regression

When Y is binary: from MLE to inference

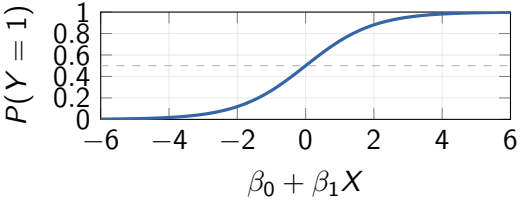
Logistic Regression as MLE

Connects to L5 (MLE) and L6 (cross-entropy)

$$Y_i \in \{0,1\}. \qquad P(Y_i = 1 \mid X_i) = \sigma(\beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip})$$

$$\text{where } \sigma(z) = \frac{1}{1+e^{-z}} \qquad \text{(logistic / sigmoid function)}$$

No closed-form solution. Fit by *maximizing the likelihood* (numerical optimization).



The negative log-likelihood is the **cross-entropy loss** from L5. MLE = minimize cross-entropy.

Inference for Logistic Regression

No exact t-distribution. Instead, use asymptotic normality of MLE (from L5):

$$Z = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \dot{\sim} N(0,1)$$

SE from the Fisher information matrix: $\text{SE} = \sqrt{[\mathcal{I}(\hat{\beta})^{-1}]_{jj}}$

Example: Predicting college admission (Y) from GPA (X_1) and test score (X_2).

	Estimate	**SE**	Z	p**-value**
(Intercept)	-12.3	2.1	-5.86	< 0.001
GPA	3.2	0.8	4.00	< 0.001
Test score	0.015	0.005	3.00	0.003

Both GPA and test score are significant predictors of admission.

Interpreting Logistic Coefficients: Odds Ratios

In logistic regression: $\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots$

A 1-unit increase in X_j multiplies the **odds** by e^{β_j}.

Odds Ratio $= \quad e^{\hat{\beta}_j}$

Admission example:

GPA: $e^{3.2} = 24.5$

Each extra GPA point multiplies admission odds by 24.5.

95% CI: $[e^{1.6}, e^{4.8}] = [5.0, 121.5]$

Test score: $e^{0.015} = 1.015$

Each extra point on the test increases odds by 1.5%.

95% CI: $[e^{0.005}, e^{0.025}] = [1.005, 1.025]$

OR > 1: increases odds. OR < 1: decreases odds. OR $= 1$: no effect.

OLS vs Logistic: Side by Side

	OLS (Linear)	**Logistic**
Response	Continuous $Y \in \mathbb{R}$	Binary $Y \in \{0, 1\}$
Model	$E[Y] = X\beta$	$\log(p/(1-p)) = X\beta$
Estimation	Closed form: $(X^\top X)^{-1} X^\top Y$	Numerical MLE (no closed form)
Distribution	t_{n-p-1} (exact)	$N(0, 1)$ (asymptotic)
SE source	$\hat{\sigma}^2 (X^\top X)^{-1}$	Fisher info $\mathcal{I}(\hat{\beta})^{-1}$
Coefficient	$\hat{\beta}_j$ = change in $E[Y]$	$e^{\hat{\beta}_j}$ = odds ratio
Overall test	F-test	LRT ($-2 \log \Lambda \sim \chi^2_p$)
Fit metric	R^2, adj. R^2	AIC, deviance, pseudo-R^2

Same workflow: Fit model $\rightarrow$ check significance of each $\hat{\beta}_j$ $\rightarrow$ check overall fit $\rightarrow$ diagnostics.
The tools change (exact t vs asymptotic Z, R^2 vs deviance), but the logic is identical.

Summary: Regression Inference

Model: $Y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$. Assumptions: LINE.

SE: $\mathrm{SE}(\hat{\beta}_j) = \hat{\sigma}\sqrt{(X^\top X)^{-1}_{jj}}$. Small SE = precise estimate.

t-test: $T = \hat{\beta}_j / \mathrm{SE}(\hat{\beta}_j) \sim t_{n-p-1}$. Tests if predictor j matters.

CI: $\hat{\beta}_j \pm t_{\alpha/2} \cdot \mathrm{SE}$. Contains 0 $\Leftrightarrow$ not significant.

Overall F: Tests all predictors at once. ANOVA for regression.

R^2: Proportion explained. Use adjusted R^2 to compare models.

Gauss–Markov: OLS is BLUE. But biased estimators (ridge) can win.

Logistic: MLE, Z-test (asymptotic), odds ratios $= e^{\hat{\beta}_j}$.

Practical: Regression Inference in Python

1. **OLS regression:**
   - Load a dataset (e.g., `sklearn.datasets.load_diabetes`)
   - Fit with `statsmodels.api.OLS`. Read the summary table
   - Identify significant predictors, interpret R^2, check overall F
2. **Diagnostics:**
   - Plot residuals vs fitted. QQ plot of residuals
   - Compute leverage and Cook's distance. Identify influential points
   - `statsmodels.stats.outliers_influence.OLSInfluence`
3. **Logistic regression:**
   - Binary outcome (e.g., `sklearn.datasets.load_breast_cancer`)
   - Fit with `statsmodels.api.Logit`. Compute odds ratios
   - Compare coefficients and CIs with OLS intuition

Homework

1. Simulated data: generate $Y = 3 + 2X_1 + 0 \cdot X_2 + \varepsilon$, $\varepsilon \sim N(0, 4)$, $n = 50$.
   (a) Fit OLS with both X_1 and X_2. Which coefficient is significant?
   (b) Compute 95% CIs. Does the CI for β_2 contain 0?
   (c) What is R^2? What is adjusted R^2?

2. Using the diabetes dataset in `sklearn`:
   (a) Fit a multiple regression. Which predictors are significant at $\alpha = 0.05$?
   (b) Plot residuals vs fitted. Is there evidence of heteroscedasticity?
   (c) Find the point with the highest Cook's distance. Does removing it change conclusions?

3. Logistic regression: Use the breast cancer dataset.
   (a) Fit a logistic model with 3 predictors of your choice.
   (b) Compute odds ratios and their 95% CIs.
   (c) Which features increase the odds of malignancy?

4. **Theory:** Explain why adding a useless predictor ($\beta_j = 0$) increases R^2 but can decrease R^2_{adj}. What does this tell us about model selection?

Recommended Resources

Video: StatQuest — Linear Regression, Clearly Explained

Step-by-step: fitting, R^2, p-values, and diagnostics. Also: "Logistic Regression."

Interactive: Seeing Theory — Regression Analysis (Brown)

`seeing-theory.brown.edu/regression-analysis` — drag points, see how the line and R^2 change.

Reading: Wasserman, "All of Statistics" — Ch. 13

Linear and logistic regression, inference, diagnostics. Concise and rigorous.

Reading: James et al., "ISLR" — Ch. 3–4

Linear regression (Ch. 3) and logistic regression (Ch. 4). Intuitive, with R/Python labs.

Python: `statsmodels`

`OLS`, `Logit`, `.summary()`, `.conf_int()`, `OLSInfluence`, `get_robustcov_results`.

Questions?

Next: Lecture 13 — Generalized linear models